

EDUCATION

Molecular Biology and Genetics, B.Sc. | Uludağ University, Bursa, Türkiye | Oct 2023 – July 2027 | GPA: 3.27/4 (Fully Funded Turkish Government Scholarship)

RESEARCH EXPERIENCE

Undergraduate Researcher

Oct 2024 – Present

Prof. Berrin Tunca | Uludağ University, Türkiye

- “Characterizing Tumor Microenvironment Transcriptome Variations in NASH and Viral Hepatocellular Carcinomas for Targeted Therapy.”
- Proposed and secured a grant to perform cross-species scRNA-seq integration and differential analysis of the poorly treated and characterized NASH-HCC, which lacks human scRNA-seq data, and validated in vitro using an FFA-induced HCC model. (TÜBİTAK 2209-A Grant Funded)
- The preliminary results poster was awarded the honor prize at a bioinformatics symposium and was later presented at an international immunology symposium.

Undergraduate Researcher

Jul 2026 – Sep 2026

Assoc. Prof. Oğün Adebali | Sabancı University, Türkiye

- Developed a phylogenomics framework to map mutational tolerance and nucleotide-level evolution rates across pathogenic bacterial genomes. This was used to identify fast-evolving regions that correlate with antimicrobial drug resistance and slow-evolving regions that contain potential therapeutic targets.
- Built a scalable Snakemake pipeline for MSA, phylogeny reconstruction, and evolution rate scoring for pathogens. (Türkiye Scholarships Funded Internship)

Bioinformatics Intern

Jan 2026 – Feb 2026

Asst. Prof. Burçak Otlı | Middle East Technical University, Türkiye

- Worked with Nextflow pipelines for large-scale WES/WGS cancer genomics, including somatic variant calling and mutational signature analysis.
- Performed basic processing and analysis of multiple omics data types at scale, including: ChIP-seq, Methyl-seq, and Bulk RNA-seq. (EU-funded SUSIA Internship)

Undergraduate Researcher

Jul 2025 – Oct 2025

Prof. Vicente Pelechano Garcia | Karolinska Institutet, Sweden

- Developed a regression-based method to score cancer cell plasticity and decode its gene signature in Hepatocellular Carcinoma (Huh7) using CITE-seq data.
- Validated the cell plasticity signature in TCGA-LIHC by Cox regression analysis, linking it to drug resistance, relapse, and survival.
- Identified distinct non-genetic phenotypic states in clonal Huh7 cell populations using scRNA-seq. (Erasmus+ Funded Traineeship)

Research Intern

Jul 2024 – Sep 2024

Asst. Prof. Atakan Ekiz | Izmir Institute Of Technology, Türkiye

- Performed bioinformatics analyses in tumor immunology using bulk/single-cell RNA-seq, TCR-seq, and microarray data to investigate immune-related lncRNAs.
- Assisted with lentiviral transduction, cell culture passaging, mouse xenograft modeling, and flow cytometry.

Undergraduate Researcher & Intern

May 2023 – Jul 2024

Asst. Prof. Elif Uz Yıldırım | Uludağ University, Türkiye

- “Comprehensive Analysis of GPRC5A Promoter Methylation Across Cancer Types.” Methly-seq pilot study poster presented at a bioinformatics symposium.
- Carried out routine molecular genetics techniques: DNA isolation, PCR, and Gel Electrophoresis.

ACADEMIC SOCIETIES & NETWORKS

Contributor & Core Team Member

Mar 2025 – Present

nf-core

- Contributed to the nf-core/scdownstream Single Cell RNA-seq pipeline by fixing runtime errors and developing doublet detection modules.
- Authored documentation and a blog post on running nf-core pipelines in Google Colab as part of the core outreach team.

Outreach & Translation Team Leader

Jan 2024 – Present

International Society for Computational Biology - Regional Student Group (RSG) Türkiye

- Organized a scRNA-seq Hackathon and a careers Q&A panel with RSG Australia and coordinated the Türkiye local site of the nf-core hackathon in 2025 and 2026.
- Directed Turkish translation of bioinformatics learning resources to increase accessibility for students in Türkiye.

Task Force Member

Oct 2024 – Present

Young European Federation of Immunological Societies (yEFIS)

- Organized a webinar exploring immunoinformatics applications across industry and academia with the Immune Collaborators workgroup.
- Conceptualized and led development of an open-access blog page to be run by the Immune Collaborators workgroup for the community to democratize technical knowledge, managing an editorial pipeline of experimental advice and career-navigation resources for early-career researchers in immunology.

Science Team Leader

Nov 2022 – Present

UluRover | Uludağ University Robotics Society

- Led R&D for an interplanetary rover competing on simulated Martian surfaces. Integrated chemical & robotic modules into an autonomous life detecting minilab.
- Helped pitch the project to multiple industry partners and secure over \$10,000 in materials and sponsorships.
- Led the science team in an additional project that won best Presentation & Pitch at the 2025 Teknofest Biotechnology Startup Idea Competition. Placed 5th/1400.

RESEARCH SKILLS & METHODS

Experimental

- **Molecular Genetics Techniques:** DNA Isolation from Blood, Restriction Digestion, PCR, Agarose Gel Electrophoresis, Bacterial Culture & Cloning

Computational

- **Programming Languages:** R, Python, Nextflow, Bash, Arduino
- **NGS Data Analysis:** Bulk & Single Cell RNA-seq, TCR-seq, CITE-seq, Microarray, WES/WGS Processing & Variant Analysis, ChIP-seq, Methyl-seq
- **Frameworks & Environments:** Slurm (Linux HPC), Conda, Docker/Singularity, Nextflow, Snakemake, R-Shiny, Git, Arduino, Raspberry Pi
- **Machine Learning:** Basic Scikit-learn and TensorFlow